

CMSC 471 Homework 2 Report

1) $h(n)$ = Hamming Distance (aka the number of mismatched letter positions between states n and y)

	Steps	Scrabble	Frequency
Each mismatch needs ≥ 1 action	An action cost is 1	The scramble value of any letter must be at least 1	Each action cost at least 1
Lower bound: $\text{mismatch} \times 1 = h(n)$	Admissible	Admissible	Admissible

2)

Word pair	Cost measure	UCS cost	UCS nodes expanded	A* cost	A* nodes expanded	Reduction
WHY $\rightarrow$ ASK	Steps	5	773	5	11	97.9%
WHY $\rightarrow$ ASK	Scrabble	9	765	9	353	53.9%
WHY $\rightarrow$ ASK	Frequency	38.86	808	38.86	755	6.6%
BEAR $\rightarrow$ DUCK	Steps	4	691	4	6	99.1%
BEAR $\rightarrow$ DUCK	Scrabble	11	2981	11	1376	53.8%
BEAR $\rightarrow$ DUCK	Frequency	38.24	1084	38.24	720	33.6%

$100 \times \frac{773 - 11}{773} = 97.93$
$100 \times \frac{765 - 353}{765} = 53.856$
$100 \times \frac{808 - 755}{808} = 6.559$
$100 \times \frac{691 - 6}{691} = 99.132$
$100 \times \frac{2981 - 1376}{2981} = 53.841$
$100 \times \frac{1084 - 720}{1084} = 33.571$

- UCS and A* always returned the same optimal cost
- The largest reductions were 97.9% and 99.1%, when the word pair WHY $\rightarrow$ ASK and BEAR $\rightarrow$ DUCK that both had the cost measure step
- WHY $\rightarrow$ ASK under the cost measure Frequency was the least informative, as it has the lowest reduction at 6.6%.
- Heuristic always guesses at least 1 cost per remaining letter. Under steps that's fine, but under both Scrabble and Frequency the real cost is almost always higher than 1, so the estimate is too low and A* has to explore more nodes to be safe.

3)

Steps: Yes steps is symmetrical every action cost 1 in both directions, reversing the path gives the same total cost

Scrabble: No it is not symmetrical as cost depends on the replacement letter which varies by direction (counter example: WHY $\rightarrow$ ASK (cost 9) | ASK $\rightarrow$ WHY (cost 14))

Frequency: No it is also not symmetrical as cost depends on the rarity of each word entered, which also varies by direction (counter example: WHY $\rightarrow$ ASK (cost 38.86)
 ASK $\rightarrow$ WHY (cost 37.76))

4)

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word u costs $1 + \text{rarity}(u)$, where the nonnegative rarity value is stored in the dictionary.

The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal. By "more informative," we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal.

Worked for 44s

The baseline heuristic $h_0(n)$ equals the Hamming distance $d(n, g)$ between the current state n and the goal g . Because every transition costs at least 1 and any feasible path from n to g must contain at least $d(n, g)$ transitions, h_0 is admissible for the frequency objective.

A strictly stronger admissible heuristic is obtained by observing that the final transition of every path enters the goal word itself. Consequently the true optimal cost $c^*(n, g)$ satisfies

$$c^*(n, g) \geq d(n, g) + r(g)$$

whenever $n \neq g$ (and equals 0 when $n = g$), where $r(g) \geq 0$ is the rarity value stored for the goal. Define

$$h_r(n) = \begin{cases} d(n, g) + r(g) & \text{if } n \neq g, \\ 0 & \text{if } n = g. \end{cases}$$

Admissibility follows at once from the inequality above. Because $r(g) \geq 0$ we also have $h_r(n) \geq h_0(n)$ for every n . Whenever the supplied dictionary assigns a positive rarity to the goal, the inequality is strict for every state that differs from the goal in at least one letter, making h_r more informative than h_0 .

(The same reasoning can be applied to any other word that is known a priori to appear on every optimal path; the goal is simply the most obvious such word.)

45 sources

Explore monotonicity constraints

Ask anything

Model used: Grok | The heuristic it proposed: $h_r(n) = \begin{cases} d(n, g) + r(g) & \text{if } n \neq g \\ 0 & \text{if } n = g \end{cases}$

Admissibility proof: Any path must enter g last, paying $1 + r(g)$, remaining spots cost ≥ 1 each, so total $\geq d(n, g) + r(g) \rightarrow$ admissible, no repair needed

Experimental result tables comparing h_0 vs h_r node counts for frequency

Problem	Heuristic	Nodes Expanded	Solution Cost
WHY $\rightarrow$ ASK	h_0	755	38.856
WHY $\rightarrow$ ASK	h_r	634	38.856
BEAR $\rightarrow$ DUCK	h_0	720	38.240
BEAR $\rightarrow$ DUCK	h_r	302	38.240

Conclusion: h_r expands way less nodes than h_0 making out pre from h_0 , even though solution cost are same

AI use disclosure: I used Claude to assist with implementing the code for doc.py. Checked the code by running it.